Chemistry - Undergraduate Level Assessment 8 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. What is the maximum number of phases that can be at equilibrium with each other in a three component mixture?
   (a) 2
   (b) 3
   (c) 4
   (d) 5

2. Which of the following is an n-type semiconductor?
   (a) Silicon
   (b) Diamond
   (c) Silicon carbide
   (d) Arsenic-doped silicon

3. Considering 0.1 M aqueous solutions of each of the following, which solution has the lowest pH?
   (a) Na_2CO_3
   (b) Na_3PO_4
   (c) Na_2S
   (d) NaCl

4. The fact that the infrared absorption frequency of deuterium chloride (DCl) is shifted from that of hydrogen chloride (HCl) is due to the differences in their
   (a) electron distribution
   (b) dipole moment
   (c) force constant
   (d) reduced mass

5. What is the NMR frequency of 31 P in a 20.0 T magnetic field?
   (a) 54.91 MHz
   (b) 239.2 MHz
   (c) 345.0 MHz
   (d) 2167 MHz

True / False (10 marks)

Answer True or False

6. True or False: The set of sp^3 hybrid orbitals for carbon is constructed from a combination of three atomic orbitals.

7. True or False: The reduction of D-xylose with NaBH 4 produces a racemic mixture of enantiomers.

8. True or False: Fluorine (F) has the lowest electron affinity due to its high electronegativity and tendency to attract electrons.

9. True or False: The thermal stability of group-14 hydrides increases in the order: PbH 4 < SnH 4 < GeH 4 < SiH 4 < CH 4.

10. True or False: The polarization of a proton is 0.793 x $10^{-}4at0.335Tand6.931x10^{-}7at10.5T$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the implications of the Heisenberg uncertainty principle for a quantum mechanical particle in the lowest energy level of a one-dimensional box, focusing on the limitations of measuring position and momentum.

12. Discuss the decay of a radioactive isotope with a half-life of 6.0 hours, starting with an initial activity of 150 mCi. Calculate the remaining activity after 24 hours and explain the underlying principles of radioactive decay.

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